StoryWritedBy

Lautaro de gyldenfeldt, Reynaldo Martín Peralta Recio, Mario sigal Aguirre
Procesamiento de Lenguaje Natural

Long Story Generation

La generación de historias extensas sigue siendo un reto para los LLM, debido principalmente a dos factores:

- Coherencia discursiva, que requiere consistencia argumental, coherencia lógica y completitud.
- Complejidad narrativa, que requiere una narrativa entrelazada y atractiva.

StoryWriter es un marco de generación de relatos multiagente, propuesto por investigadores de la universidad de Tsinghua, Beijing, China, para abordar esto.

Author Imitation

La tarea de *author imitation* busca generar textos originales que reproduzcan el estilo característico de un autor determinado —incluyendo su sintaxis, ritmo narrativo, elección léxica, estructura discursiva y patrones recurrentes— sin copiar obras preexistentes. En modelos de lenguaje modernos entrenados a gran escala, esta capacidad emerge parcialmente en configuración *zero-shot*, pero en este proyecto nos enfocamos en **Fine-Tuning específico por autor** que es otra de las maneras más estudiadas para implementar este tipo de capacidad a modelos.

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Nuestro proyecto, se basa en la conjunción de estas dos últimas definiciones, proponemos un marco de generación de relatos, inspirados en los de un autor en específico, basado en un sistema multiagente reajustado utilizando obras de este autor. También mediremos la performance de este reajuste, comparándolo con el modelo original en configuración *zero-shot.*

Arquitectura del Sistema

Tanto StoryWriter como nuestra propia implementación, constan de tres módulos principales:

- Agentes de esquema: generan un esquema para cada historia, formados por eventos que contienen tramas detalladas, personajes y relaciones entre eventos.
- Agentes de planeación: detallan los eventos y planifican qué eventos deberán escribirse en cada capítulo para mantener una historia entrelazada y atractiva.
- Agentes de escritura: generan cada capítulo basándose en la historia creada hasta el momento y el evento actual, asegurándose de mantener coherente la historia completa.

Descripción de cada Agente

Agentes de esquema:

1. El agente *EventSeed* es el encargado de generar eventos secuencialmente, basándose en la premisa proporcionada por el usuario, construye el esquema de la historia proporcionando información esencial para cada evento, como puede ser la hora, ubicación, relaciones, etcétera.
2. El agente *EventValidator* se encarga de monitorear y evaluar continuamente los outputs generados por *EventSeed*, proporcionando un feedback para mantener coherencia entre eventos y guiar la generación de los eventos subsiguientes.

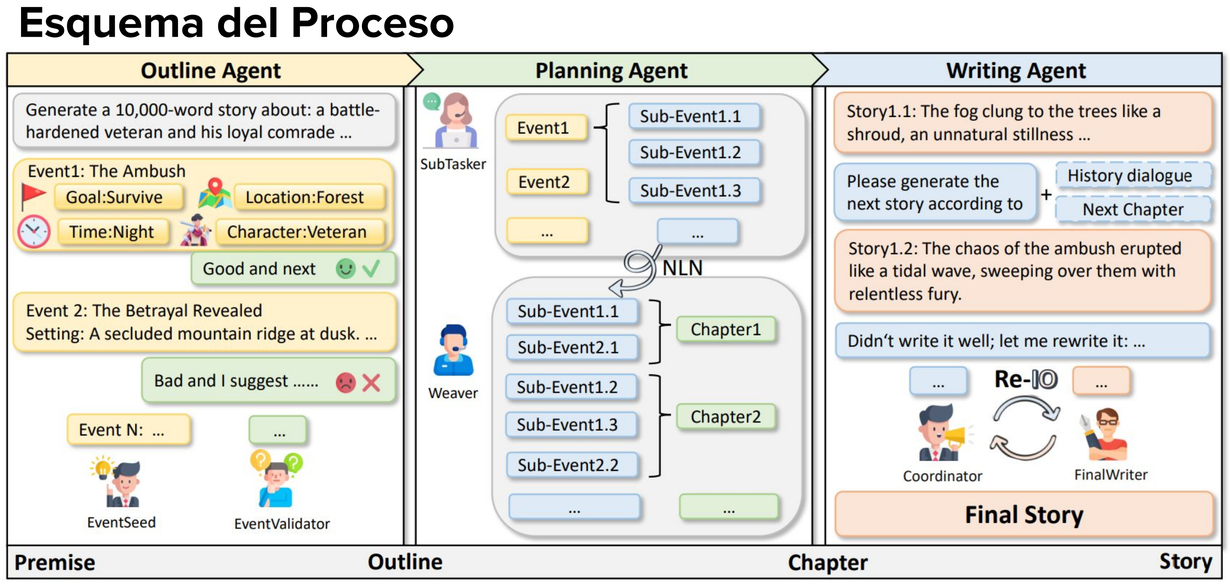
Agentes de planeación:

3. El agente *SubTasker* se encarga de generar subeventos descomponiendo eventos de alto nivel en unidades narrativas más detalladas.
4. El agente *Weaver* asigna estos subeventos a diferentes capítulos, garantizando la coherencia de la estructura narrativa general y permitiendo una presentación no lineal

Agentes de escritura:

5. El agente *FinalWriter* se dedica a general la narrativa final, con especial énfasis en garantizar la uniformidad estilística y una alta calidad textual
6. El agente *Coordinator* asume la responsabilidad de supervisar la arquitectura narrativa global, pudiendo aprobar o solicitar reescribir el capítulo generado por el agente *FinalWriter*.

Esquema del Proceso



Alineación del Modelo al Estilo del Autor

Se eligió como modelo base a **Mistral** con 7B de parámetros en su versión *Instruct*, adaptada para seguir instrucciones humanas. A su vez, el autor elegido para representar fue **William Shakespeare**, uno de los escritores más influyentes de la literatura inglesa, posiblemente el más destacado de todos.

Para realizar la adaptación del modelo al estilo literario del autor se empleó **LoRA** (*Low-Rank Adaptation*) una técnica *fine-tuning* que introduce matrices de bajo rango en algunas capas del modelo y mantiene congelados los pesos originales del mismo, lo que permite entrenar únicamente este subconjunto de parámetros adicionales, reduciendo el costo computacional.

En este trabajo se utilizó una **configuración** con una *dimensionalidad de bajo rango* r = 32 y un *factor de escalamiento* α = 32, junto con un *dropout* de 0.05. No se entrenaron parámetros de *bias*.

Está configuración mencionada se aplicó sobre las capas de *atención* y los bloques *MLP* del *transformer*, que son los componentes más influyentes para la generación de texto del modelo, resultando en un total de 84 millones de parámetros entrenables.

El **entrenamiento** se llevó a cabo durante 2 *epochs*, con un *batch size* de 8 instancias y un *learning rate* de 2e-5 con *cosine decay* y *warmup* del 3%. El optimizador utilizado fue *Paged AdamW 8-bit*.

División de Datos

Los **datos** utilizados en el entrenamiento provienen del libro *The Complete Works of William Shakespeare* (en adelante, *TCWWS*), descargado desde el sitio oficial de Project Gutenberg, junto con una colección adicional de obras de otros autores empleadas posteriormente para tareas de evaluación. El libro *TCWWS* fue sometido a un proceso de limpieza y luego segmentado en 3000 particiones:

- 2400 se destinaron al proceso de *fine-tuning* del modelo,
- 560 al entrenamiento y validación de un *clasificador* (subconjunto *Control Positivo*), y
- 40 al conjunto de *benchmark*.

De forma adicional, los libros de otros autores también fueron limpiados y segmentados, y luego organizados en dos subconjuntos:

1. *Control Negativo Grueso*: conformado por autores de distintas épocas, estilos y géneros literarios. Este grupo incluyó 620 particiones, de las cuales 600 se utilizaron para entrenar el *clasificador* y 20 para el *benchmark*.
2. *Control Negativo Fino*: integrado por autores contemporáneos a *Shakespeare* y/o pertenecientes a géneros similares. Este conjunto estuvo compuesto por 120 particiones, con 100 asignadas al entrenamiento del *clasificador* y 20 al *benchmark*.

Evaluación de Author Imitation

Se entrenaron dos modelos clasificadores, basado en **Roberta**, con los datos mencionados anteriormente, dedicados a decidir si un texto mantiene el estilo de *William Shakespeare*. Uno de estos modelos (Roberta A) solo utiliza *Control Positivo* y *Control Negativo Grueso*, y el otro (Roberta B) agrega también *Control Negativo Fino*.

Posteriormente, se midió el **TPR** (True Positive Rate, $P(\hat{Y}=+ \mid Y=+)$) y **FPR** (False Positive Rate, $P(\hat{Y}=+ \mid Y=-)$) en ambos modelos y en base a eso y la predicción del modelo frente a los textos generados, se pudo obtener un **score** π de cada modelo. Los resultados se muestran en la Tabla 1 y Tabla 2.

A su vez, también se midió la Perplexity del modelo sobre los conjuntos de control. Resultados en Tabla 3.

$$\pi = \frac{r - f}{t - f}$$ Con r= CasosPredichosPositivos/CaosTotales, t=TPR, f = FPR

Resultados Experimentales

Se generó una historia de 5 capítulos utilizando como agente de *FinalWriter* al modelo *Mistral Fine-Tuned*. los resultados se muestran a continuación.

CHAPTER I — *Upon the Threshold of Misfortune*

In a land where dawn seemed forever reluctant—hovering betwixt its birth and a timid retreat into the bosom of night—there wandered young Edric, heir to no crown, sworn to no banner, and pilgrim to no foretold destiny. His attire bore the dust of many roads, his boots the weary testament of unnumbered miles; yet despite the plainness of his men, there stirred within him a quiet nobility, born not of lineage but of a heart ungoverned by fear. Indeed, even the wind, gliding past his solitary form, seemed to incline itself in subtle reverence, as though acknowledging a truth he himself had not yet glimpsed.
For many nights he had journeyed without the comfort of direction, guided only by a restlessness he could neither name nor quell. And now, at last, his wandering brought him to the borders of Ormere Forest, that ancient expanse which the villagers of nearby hamlets spoke of in hushed tones, uncertain whether its name summoned wonder or calamity. Tales abounded of spirits that warred in silence, of roots that remembered the weight of forgotten kings, and of shadows that moved with purpose unbecoming of mere darkness.
Edric halted at the forest's edge.
Towering boughs rose before him like pillars of some cathedral wrought not by mortal hand but by the whims of eternity itself. The leaves, trembling upon threads of silvered light, whispered secrets too ancient for mortal comprehension. Though the morning sun labored to pierce the canopy, its rays faltered and broke upon the dense foliage, as though even light itself dared not intrude.
A curious stillness seized the air.
The hum of insects quieted.
The rustle of small creatures fled.
Even the breeze that moments before had played upon Edric's cloak now retreated, as though unwilling to witness the next turn of fate.
"Strange," he murmured, his voice swallowed almost instantly by the heavy hush.
"As though the very world holds its breath."
His hand brushed the hilt of the modest blade at his side—a tool more suited to foraging than battle. He felt no safer for it. And yet, something within him stirred with the faintest thrill of recognition, as though unseen eyes called him forward, not in threat, but in expectation.
He stepped into the shadow.
At once, the forest seemed to shift. The air thickened; sounds muffled; the ground beneath his boots felt firmer, as though roots coiled just beneath the surface, testing the presence of an intruder. The trees bent subtly inward, arching over him like ancient guardians lifting their solemn watch.
With each pace he took, the world behind him faded—distant, irrelevant—while the world before him deepened into a labyrinth of dim-lit paths and murmuring branches. Shadows elongated, some stretching in ways that did not obey the angle of the sun; others clung to hollows as though nursing dormant memories.
Edric's breath grew slow and deliberate.
For though he had heard a lifetime of tales woven around Ormere's darkened heart, nothing in those whispered stories compared to the palpable gravity that now pulled upon his spirit. It was as if fate, long idle and patient, had risen from its slumber to lay its hand upon his shoulder.
And in that moment, a truth ascended from the marrow of the earth:
This venture was not his choosing. It had been waiting—centuries waiting—for him.
Yet no prophecy heralded his arrival. No celestial sign split the heavens to announce his first steps. No sage proclaimed him chosen.
His journey began instead in silence.
A single step, then another, into the murmuring dark—where destinies forgotten by all but the forest itself lay waiting to awaken, eager to claim the soul of any who dared disturb their ancient repose.
Thus commenced the path of Edric: not with fanfare, nor divine portent, but with quiet defiance beneath ancient boughs, where fate threads its tapestry with invisible strands—perilous, curious, and absolute.

CHAPTER II — *Where Shadows Speak of Oaths Forgotten*

The forest received Edric as a somber host might welcome an unexpected guest: with courtesy veiled beneath suspicion. Each bough arched overhead like the ribbed vault of some ancient cathedral, and the dim light filtered through in trembling shards, as though reluctant to declare allegiance either to day or to darkness.
Edric's boots sank softly into the moss, which yielded under his weight like a memory long suppressed. No creature stirred—no bird dared break the hush—yet he felt not alone. A presence, subtle as an echo and persistent as remorse, walked beside him unseen.
"Who goes there?" he called, though scarcely above a whisper, for the silence of Ormere seemed a thing alive, easily offended.
No voice replied; yet from the hollow of a twisted oak emerged a faint glow, wavering like a flame half-remembered. Drawn to it, Edric approached, and beheld upon the bark an inscription worn by centuries:
"Swear upon the wandering star,
Lest thou be lost to who thou art."
He traced the runes with trembling fingers. A chill coursed through him, not of fear alone but of recognition—a stirring of some destiny etched into his being before his first breath.
As he pondered the verse, a gust rose, swirling the leaves in a frantic dance. From the heart of that tempest formed a figure cloaked in shadow, its outline wavering like smoke.
"Traveler," it intoned, with a voice neither man's nor spirit's, "the star seeks one who remembers the oath once broken. Speak thy purpose, lest the forest judge thee unworthy."
Edric steadied himself, though his pulse thundered like a war drum.
"I seek the truth of my birth," he declared. "And whatever chains it forged upon my fate."
A pause—heavy, measuring, ancient.
"Then tread lightly," the shadow warned. "For truths buried in Ormere seldom rise without demanding a price."
And as the figure dissolved into the swirling dark, Edric sensed that the forest had accepted his answer—but not without expectation. The oath of the wandering star had stirred once more, and the debt it called for would not be small

CHAPTER III — *The Star's Whisper in the Gloaming*

Long after the spectral figure had faded into the restless dark, Edric remained rooted to the spot, as though the very earth clutched at his ankles. The forest, relieved of its momentary frenzy, settled back into a silence too complete to be natural. It pressed upon him like a shroud, soft yet inexorable.
At length, he forced his legs onward, guided only by a frail ribbon of light that wove between the trees—a pale shimmer that flickered with each beat of his uncertain heart. Though he knew not whether it urged him toward destiny or ruin, he felt somehow compelled to follow it, as though his very soul had been tethered to its glow.
The path narrowed, then widened again, revealing a clearing bathed in twilight. There, upon a stone worn smoother than the memory of its own creation, lay a small silver orb. It pulsed faintly, each throb like the sigh of a distant star.
Edric approached with reverence.
The orb quivered, and from within it rose a whisper—thin, melodic, unmistakably beckoning.
"Edric… child of the forgotten vow…"
He jerked back, startled. "Who speaks? Show thy face, or release me from this torment!"
The orb brightened, lifting from the stone as though freed from gravity's hold. Its light spilled across the clearing, illuminating a mural carved upon the surrounding rocks: a figure crowned in starlight, extending a hand toward a kneeling supplicant.
Beneath the carved figures, he read words nearly erased by time:
"One shall rise,
Bound by oath and severed ties."
A tremor began in Edric's chest—fear, yes, but something deeper. Recognition. As though these verses, these omens, belonged to him as surely as breath or bone.
The whisper returned, firmer now, threaded with urgency.
"Seek the Mirror of Aethryn in the heart of Ormere. Only there shall the truth unveil—and the debt of the oath be reckoned."
Edric swallowed hard. "And if I refuse?"
The light dimmed, flickering like a candle in its last breath. Then, in a tone mournful as winter wind, came the answer:
"Refusal is but another path to the same end. The star has chosen. Its judgment cannot be undone."
With that, the orb fell silent, settling once more upon the ancient stone. Edric, shaken yet resolved, stepped closer and lifted it into his hands. It was warm—warm as life, warm as fate.
He turned toward the deeper woods, where shadows stretched like claws and the faintest trail wound into obscurity.
Whatever awaited him—truth, ruin, or revelation—it had already begun to weave itself around him.
And thus, with the star's whisper echoing in his mind, Edric walked into the gloaming, toward the Mirror of Aethryn and whatever destiny dared await him there.

CHAPTER IV — *The Mirror's Path and the Burden of the Oath*

The forest grew darker as Edric pressed deeper into its ancient heart. Not the darkness of night, but a gloom woven from age and memory—a dimness that seemed to cling to the bark of trees and trail behind him like a reluctant companion. Each step stirred murmurs in the undergrowth, as though unseen watchers debated his worth in hushed tones.
The silver orb in his hand pulsed gently, guiding him toward a narrow path veiled beneath arching boughs. No birds called. No breeze stirred. Even Edric's breath felt subdued, as if the air itself feared to break the solemnity of his passage.
At last, the trail broadened into a cavern of trees whose trunks twisted skyward like pillars of a forgotten temple. At the center stood an obelisk of obsidian, its surface polished to a mirror-like sheen despite the encroaching moss.
Edric felt his heartbeat falter.
"The Mirror of Aethryn…" he breathed.
As he approached, the orb warmed, slipping from his grasp to float before the dark monolith. Its light washed over the surface—and the obsidian awakened. Ripples shimmered across it, as though he gazed into a pool of ink stirred by an unseen hand.
From the shifting depths emerged a reflection—but not his.
A man stood within the glass, cloaked in starlit silver, bearing a crown wrought from the constellations themselves. His gaze was sharp, sorrowful, regal. The very air trembled beneath its weight.
Edric staggered back. "Who… who art thou?"
The figure's voice resonated like distant thunder softened by grief.
"I am Aethryn, King of the Wandering Star. And thou, Edric, art bound to me by a vow sworn before thy first cry met the world."
A chill seized Edric's spine. "A vow? I know of no such oath."
"Yet it knows thee."
The reflection shifted, revealing an infant swaddled in silver cloth, placed upon the stone where Edric now stood. The starlit king knelt, touching the child's brow with a trembling hand.
"In an age of ruin, I bound my power to the blood of one yet unborn, that hope might not perish with my failing reign. Thy lineage was chosen. From thee would rise the Keeper of the Star—a guardian to mend what I could not."
A tremor began in Edric's chest—fear, yes, but something deeper. Recognition. As though these verses, these omens, belonged to him as surely as breath or bone.
The whisper returned, firmer now, threaded with urgency.
"Seek the Mirror of Aethryn in the heart of Ormere. Only there shall the truth unveil—and the debt of the oath be reckoned."
Edric swallowed hard. "And if I refuse?"
The light dimmed, flickering like a candle in its last breath. Then, in a tone mournful as winter wind, came the answer:
"If its Keeper fails to rise, the world shall follow it into darkness."
Edric clenched his fists. Fear warred with disbelief, yet beneath both flickered a spark—one he had long ignored. A feeling that he had always been meant for something he could neither name nor understand.
"What must I do?"
The king raised his hand. The orb glowed brightly, aligning itself with Edric's heart.
"Take the star's light into thy spirit. Seek the Sanctuary of the Fallen Sky. There the oath shall complete—and thy fate shall be sealed."
The forest trembled. The mirror dimmed. The orb drifted back into Edric's grasp, blazing with purpose.
Suddenly, a howl split the air—deep, ragged, ancient. The trees shuddered. Edric spun around.
From the shadows emerged eyes like burning coals.
The guardians of the oath had awakened.
And they were not eager to let their chosen escape untested.

CHAPTER V — *Where the Star Chooses Its Last Keeper*

The forest had grown still—eerily, impossibly still—as though all of Ormere held its breath for the ending of a tale begun long before Edric ever drew breath. The silver orb in his hand, once a timid flicker of destiny, now blazed with the certainty of a rising sun. Its warmth no longer trembled; it surged.
Behind him, the Mirror of Aethryn had dimmed to dull stone. Before him, the clearing stretched wide and solemn, the air heavy with a hush that felt like a prayer awaiting its final word.
Then came the howl.
Not the cry of warning he had heard before, but a mournful call—low, resonant, almost… grateful.
The Hounds of the Oath appeared one last time, their forms no longer sharp nor threatening, but frayed at the edges, dissolving into currents of stardust. Their task neared its end. The oath they guarded trembled on the edge of fulfillment.
The largest of them approached Edric, its eyes no longer embers but gentle coals nearing their final glow.
"Keeper." The word came soft as falling ash.Edric swallowed. "My trial… is done?"The hound bowed its head.
"The world turns on thy choosing."
The orb pulsed, brighter and brighter, until its light spilled onto the treetops, splitting the gloom with a radiance that felt both triumphant and sorrowful. Edric lifted it instinctively—and the forest answered.
Branches bent low in reverence. Leaves shivered like chimes. The very ground hummed with ancient recognition.
Within the orb's heart, a familiar silhouette formed: Aethryn, the starlit king, his figure faint but serene.
"Child of my last hope," he said, voice thin as wind through broken stone. "The star wanes. Only one act remains to mend what centuries unmade."
Edric felt a tremor in his chest. "If I accept the oath fully… what becomes of me?"
The king's sorrow smiled through the fading light.
"Thou shalt not perish… yet thou shalt no longer walk among men."
"The Keeper's spirit joins the star, that its light may shine where no mortal may tread."
The words struck like a bell of fate tolling its final stroke.He understood. Perfectly. Painfully.He would save the world— but never again belong to it.The hound raised its head, its voice barely a whisper:"Choose." Edric looked upon the clearing— where he had first trembled, first faltered, first dared to believe he was more than a wanderer.
He thought of the villages beyond Ormere, of children who still laughed, of dawns that deserved to rise without fear of the dark creeping over them.He did not hesitate.He lifted the orb high."I accept." Light erupted.
A great wind swept through the forest, scattering starlit dust like snow made of forgotten dreams. The hounds lifted their heads, howling in a harmony both mournful and triumphant.
The trees shone from within, as if the star's heart had been poured into every root and leaf.
Edric's body dissolved into radiance— first his hands, then his form, then the very shadow he cast— until all that remained was a pillar of light ascending toward the trembling heavens.
The orb shattered into a thousand gleaming shards.
And from the sky above Ormere, a new star ignited. Small at first.Then brilliant. Then unwavering.
The hounds watched in silence as their final Keeper took his place among the constellations.
The largest among them bowed deeply.
"Rest, Edric of the Oath," it murmured.
"Thy light shall guard us long after our names are dust."
The wind stilled. The forest sighed.
And far above, where the night's tapestry stretched endless and serene, a lone star gleamed with a warmth unmistakably human— the last gift of a wanderer who had found, at last, both purpose and eternity.

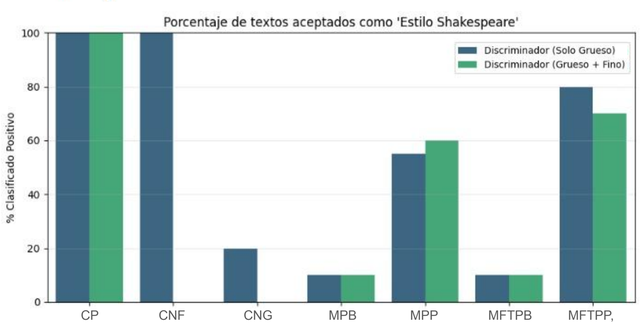


Tabla 1: Resultados de las predicciones positivas obtenidas del modelo clasificador. CP, CNF, CNG, MPB, MPP, MFTPB, MFTPP representan los dataset *Control Positivo*, *Control Negativo Fino*, *Control Negativo Grueso* y los textos generados por los modelos *Mistral con Prompt Base*, *Mistral con Prompt Perfeccionado*, *Mistral Fine-Tuned con Prompt Base*, *Mistral Fine-Tuned con Prompt Perfeccionado* respectivamente.

Clasificador	MPB	MPP	MFTPB	MFTPP	TPR	FPR
Roberta A	-0.71	0.14	-0.71	0.61	1	0.47
Roberta B	0.1	0.6	0.1	0.7	1	0

Tabla 2: Resultados de los scores obtenidos en la clasificación de textos. MPB, MPP, MFTPB, MFTPP, TPR, FPR representan los modelos *Mistral con Prompt Base*, *Mistral con Prompt Perfeccionado*, *Mistral Fine-Tuned con Prompt Base*, *Mistral Fine-Tuned con Prompt Perfeccionado* y los valores de *True Positive Rate* y *False Positive Rate* respectivamente.

Dataset de Prueba	PPL Modelo Base	PPL Modelo Fine-Tuned
Shakespeare	76.19	31.27
Corpus General (wiki)	38.13	37.19

Tabla 3: Resultados de las Perplexities obtenidas para el modelo base y el modelo finetuneado, utilizando el corpus de Shakespeare y un corpus general obtenido de Wikipedia.

Conclusión

Podemos observar que el modelo **Mistral Fine-Tuned** supera en desempeño al modelo base, lo cual evidencia que aplicar *fine-tuning* a LLMs es una estrategia efectiva para dotarlos de capacidades de *author imitation*. No obstante, la implementación práctica revela ciertas limitaciones: el proceso resulta complejo, costoso y poco escalable cuando se desea extender la metodología a múltiples autores.

Líneas futuras de trabajo podrían incluir comparaciones sistemáticas entre modelos medianos fine-tuneados y modelos de mayor tamaño evaluados en *zero-shot author imitation*, lo que permitiría determinar con mayor claridad el costo–beneficio de cada enfoque y delimitar en qué escenarios conviene aplicar fine-tuning y en cuáles no.